

Aerospace labor planning

Forecast workload. Find skill bottlenecks. Compare practical recovery options.

PROJECT SUMMARY

Built an Excel planning model linking 2,400 work packages, 5,475 labor entries and 75 technicians. The model combines weekly incoming workload forecasts, job-level estimate calibration, skill capacity and recovery scenarios.

4,856 h

Scenario workload
Forecast plus opening backlog

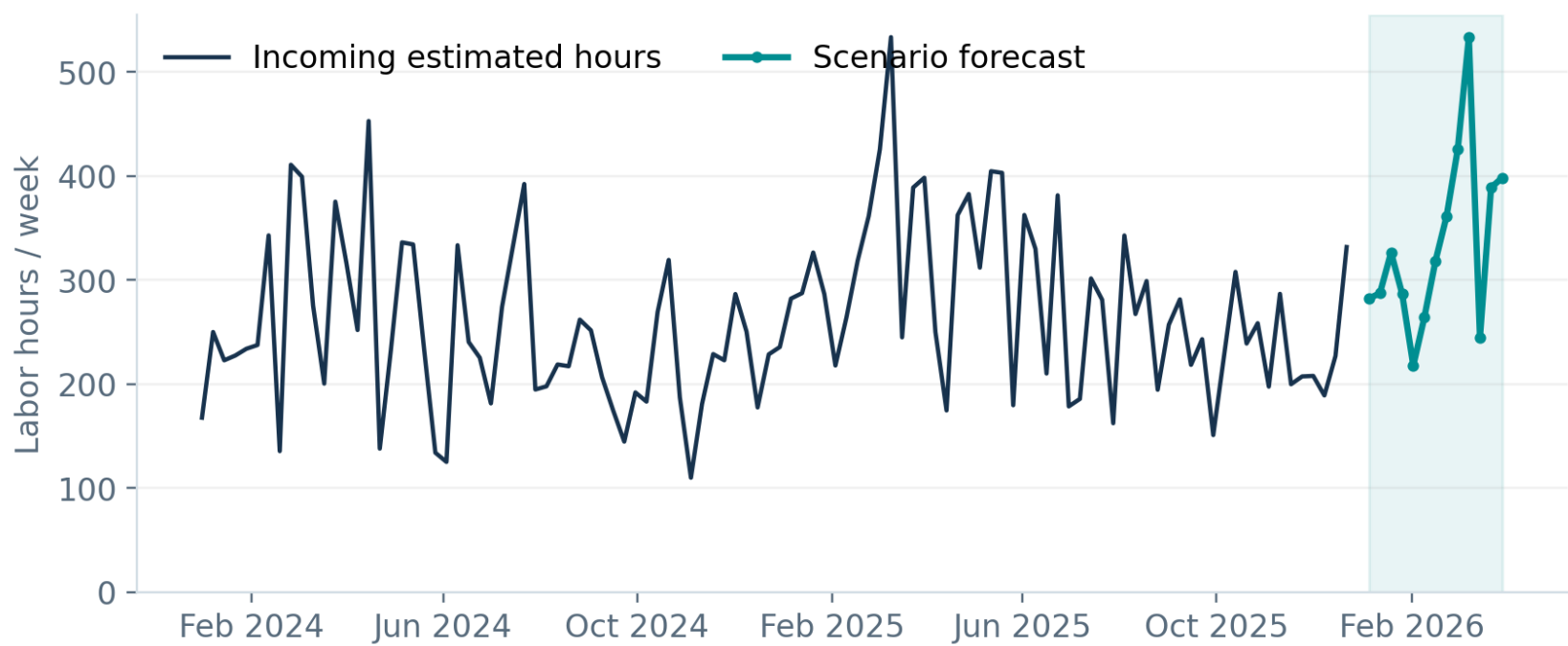
104.2%

Overall capacity utilization
Across the 13-week window

26.7%

Independent forecast error
WMAPE on 13 final-test weeks

WEEKLY WORKLOAD AND SCENARIO FORECAST



Planning implication: Aggregate capacity is tight, and Avionics reaches 120.5% utilization. A skill-specific plan is more useful than a single total labor number.

Data history: Jan 2024-Dec 2025. Scenario forecast: Jan 5-Mar 30, 2026. The final partial history week is excluded from forecast training and testing.

Synthetic data. Staffing, costs and interventions are hypothetical. Not affiliated with any aircraft manufacturer.

Test before trusting

Separate model selection from the final test, and use only information available at each cutoff.

78 weeks

Initial training

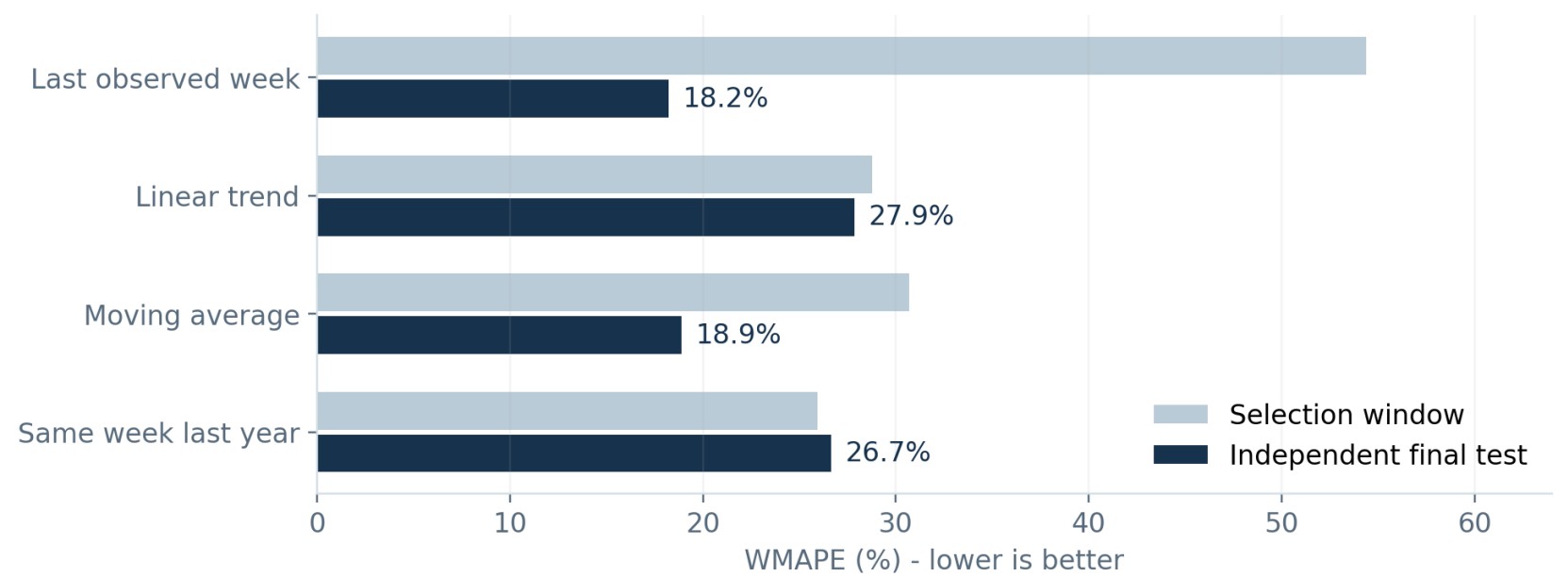
13 weeks

Method selection

13 weeks

Independent final test

FOUR METHODS, TWO SEPARATE EVALUATION WINDOWS



Result: Same week last year won the selection window (25.9% WMAPE), but scored 26.7% on the final test. Last observed week performed better on that final test (18.2%). This signals selection instability; it does not justify choosing a new winner after seeing the test.

JOB-LEVEL ESTIMATE CALIBRATION

Holdout bias improved from -7.5% to -0.6%, reducing systematic underestimation. MAE fell 4.1% (1.754 to 1.682 hours per job). Task adjustments use only jobs completed before each cutoff.

WMAPE = total absolute forecast error / total incoming estimated hours. Weekly demand means initial estimated hours at intake, not final actual labor. Forecast bands in the workbook are illustrative, not validated confidence intervals.

The bottleneck is the skill

Compare incoming workload plus opening backlog with useful capacity under the staffing scenario.

4,662 h

Allocated useful capacity

11 assigned FTE across 6 skills

3 of 6

Skills above 100% utilization

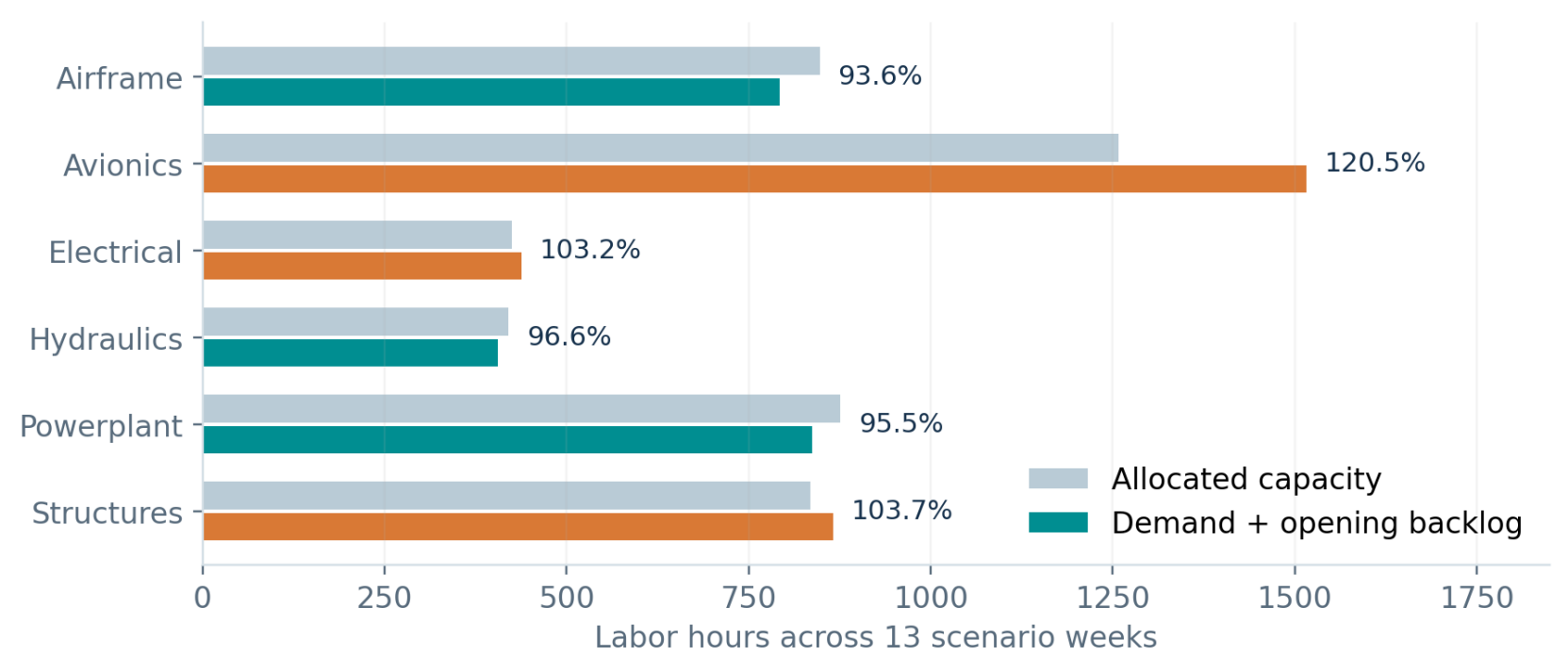
Avionics, Electrical, Structures

258 h

Avionics baseline gap

Largest skill-level shortfall

13-WEEK DEMAND VS. ALLOCATED CAPACITY



Decision insight: Avionics has the largest gap: 1,515 demand hours versus 1,258 capacity hours. Electrical and Structures also exceed their allocated capacity. The other skills still run above 90% utilization.

HOW THE SCENARIO IS BUILT

Forecast hours are split by each skill's share of incoming estimates in the last 26 complete weeks. Opening backlog is added, then demand is compared with allocated useful hours.

Assumptions: the Dec 30 snapshot backlog carries into Jan 5 without intervening clearance. Staffing and productive capacity are scenario inputs. Utilization is a workload-to-capacity ratio, not measured employee performance.

Make trade-offs visible

Stress-test Avionics with a 25% demand surge plus 120 additional unplanned hours.

716 h

Stressed capacity gap

1,974 demand - 1,258 capacity

189 h

Overtime gap reduction

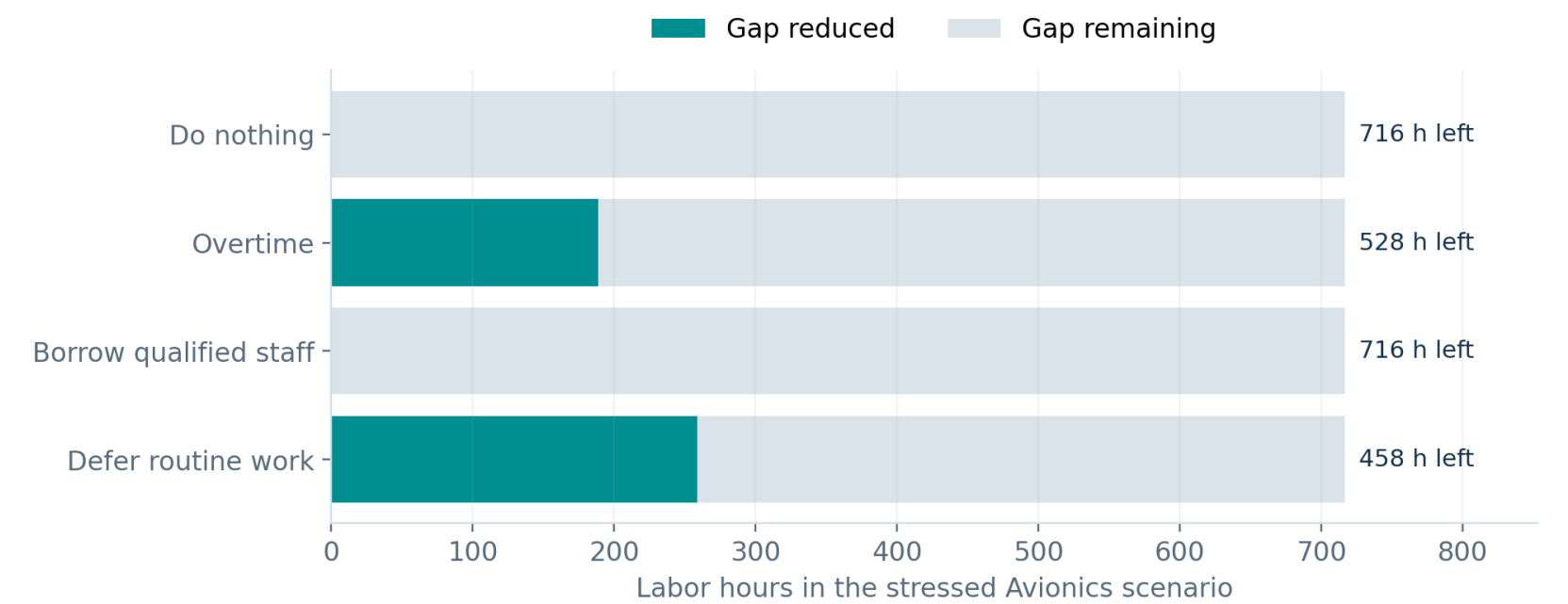
Approx. \$15,565 scenario cost

259 h

Routine hours deferred

Approx. 29 jobs move out

NO SINGLE OPTION CLOSES THE STRESSED GAP



Recovery insight: Overtime adds cost to a workforce already booking 13% overtime. Deferral moves work outside the window rather than creating capacity. Borrowing contributes zero hours under the 85% lending-skill utilization ceiling.

WHAT THIS PROJECT DEMONSTRATES

Excel formulas and linked tables; labor estimate calibration; time-based validation; skill-level capacity planning; backlog analysis; and quantified recovery trade-offs.

Before operational use: confirm task qualifications, staff availability and schedule dependencies. Borrowing assumes qualified staff with no training lead time. Costs include hypothetical overtime and training only; deferral requires program approval.